PELL, PRIME FACTORS AND A SHORTER PROOF

Pell's equation: a sharper sieve

Start with nearly equal squares; use their arithmetic to rule out triangles before calculating them.

1 / A ONE-UNIT DIFFERENCE

The first Pell equation

Look for integers with $x^2 - 2y^2 = 1$:

$$3^2 - 2 \cdot 2^2 = 1, \qquad 17^2 - 2 \cdot 12^2 = 1.$$

One square exceeds twice another by just one unit. The ratios x/y give increasingly close approximations to $\sqrt{2}$:

x	y	x/y
3	2	1.5
17	12	1.416666…
99	70	1.414285…

Indeed, $x/y = \sqrt{2 + 1/y^2} > \sqrt{2}$.

2 / MULTIPLY A NUMBER BY ITS CONJUGATE

Why the solutions keep coming

The conjugate of $x + y\sqrt{2}$ is $x - y\sqrt{2}$. Define

$$\mathrm{Nm}(x + y\sqrt{2}) = (x + y\sqrt{2})(x - y\sqrt{2}) = x^2 - 2y^2.$$

Conjugates multiply in matching pairs, so $\mathrm{Nm}(\alpha\beta) = \mathrm{Nm}(\alpha)\,\mathrm{Nm}(\beta)$. Since $\mathrm{Nm}(3 + 2\sqrt{2}) = 1$, multiplication by $3 + 2\sqrt{2}$ gives

$$\boxed{x' = 3x + 4y, \qquad y' = 2x + 3y.}$$

Starting at $(1, 0)$ produces $(3, 2), (17, 12), (99, 70), \ldots$. The same step preserves any fixed value M in the **generalised Pell equation** $x^2 - 2y^2 = M$.

3 / BRING THE ANNULUS BACK INTO VIEW

Euclid's formula supplies a Pell norm and a positive companion

A primitive right triangle has legs $m^2 - n^2$, $2mn$ and hypotenuse $c = m^2 + n^2$, with $m > n \geq 1$, $\gcd(m, n) = 1$ and $m - n$ odd. Its circumradius is $R = c/2$ and inradius $r = n(m - n)$. Thus its annulus area is

$$A = \pi(R^2 - r^2) = \frac{\pi}{4}N, \qquad N = uv, \qquad \boxed{u = c - 2r = (m - n)^2 + 2n^2, \quad v = c + 2r = (m + n)^2 - 2n^2.}$$

For primitive inputs, u, v are odd and coprime: a common odd prime would have to divide both m and n. These forms restrict the prime factors *before* we enumerate the possible pairs $uv = N$.

4 / A PRIME CAN BE FORCED TO ONE SIDE

Which primes allow 2 or -2 as a square?

If an odd prime p divides u, then $(m - n)^2 \equiv -2n^2 \pmod{p}$. If it divides v, then $(m + n)^2 \equiv 2n^2 \pmod{p}$. Here $p \nmid n$, so dividing by n^2 is allowed.

The standard square-mod-prime test says that -2 is a square for $p \equiv 1, 3 \pmod{8}$, and 2 for $p \equiv 1, 7 \pmod{8}$. Consequently:

$p \bmod 8$	allowed factor
1	u or v
3	u only
5	neither
7	v only

Coprimality puts each whole prime power on one side. A prime can change sides between triangles only if it is $1 \pmod{8}$. For $N = 21$, this already forces $(u, v) = (3, 7)$. The final square test is still needed.

5 / REFINE PAIR SEARCH

Only three primes can choose a side

$$N_0 = 87\,110\,147\,844\,489 = (3 \cdot 11)(7 \cdot 4583)(401 \cdot 449 \cdot 457).$$

The first bracket must enter u; the second must enter v. Only the last three primes are $1 \pmod{8}$. Write

$$Q = 401 \cdot 449 \cdot 457, \qquad u = 33t, \quad v = 32081\,Q/t, \quad t \mid Q.$$

There are $2^3 = 8$ allocations. Put $d = b - a$ for ordered legs $a < b$; Pythagoras gives

$$d^2 = D = \frac{u^2 + 2uv - v^2}{2}.$$

Require $u < v$ and $D \geq 0$, equivalently $1 < v/u \leq 1 + \sqrt{2}$. Testing the eight divisors of Q:

t	result
$1,\ 401,\ 449,\ 457,\ 401 \cdot 449$	$v/u > 1 + \sqrt{2}$
$401 \cdot 457,\quad 449 \cdot 457$	keep both
Q	$u > v$

6 / VERIFY THE TWO SURVIVORS

Two square checks complete this fixed-area proof

t	u	v	$D = d^2$
$401 \cdot 457$	6 047 481	14 404 369	$(1\,285\,783)^2$
$449 \cdot 457$	6 771 369	12 864 481	$(5\,223\,833)^2$

Both pairs satisfy $u \equiv v \pmod{4}$ and $\gcd(u, v) = 1$. Recover $a = (v - d)/2$, $b = (v + d)/2$, $c = (u + v)/2$:

$$(6\,559\,293,\ 7\,845\,076,\ 10\,225\,925), \qquad (3\,820\,324,\ 9\,044\,157,\ 9\,817\,925).$$

$$64 \text{ factor pairs} \longrightarrow 8 \text{ allocations} \longrightarrow 2 \text{ candidates} \longrightarrow 2 \text{ primitive triangles.}$$

There is no third primitive triangle for this N_0, with no bound on its hypotenuse.

What Pell contributes. The two quadratic forms provide an early prime filter. A Pell recurrence preserves one norm; fixing an annulus requires the product $uv = N$ and the triangle conditions as well.
The argument proves exactly two for N_0; a universal bound of two remains unproved here.

Sources and links. K. Conrad, *Pell's Equation, I*, and *Square Patterns and Infinitely Many Primes*. Companions: the annulus proof, quadratic reciprocity, and the elliptic-curve sheet.

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